

The application of large language models in coaching: A knowledge-based framework for AI-Driven coaching

Lixian Xie & Erek J. Ostrowski

Coaching is the practice of facilitating individuals' learning, growth, performance, and wellbeing. The rapid advancement of artificial intelligence (AI), particularly through the development of large language models (LLMs), has demonstrated remarkable potential across various fields, including coaching. This study investigates the transformative role of LLMs in coaching. It investigates individuals' receptivity to AI-driven coaching through a survey involving 49 university students. Results indicate a strong openness to AI coaching, with over 50% of participants expressing a willingness to use such tools while also raising concerns about privacy, effectiveness, and cultural understanding. This paper also proposes a knowledge-based framework for AI-driven coaching rooted in prominent psychological theories and coaching models. Through a structured three-step process, the framework enables LLMs to emulate professional coaching practices by leveraging prompt engineering and psychological insights to deliver personalised, accessible, cost-effective, and scalable coaching. This study underscores the significant potential of AI-driven coaching to promote human wellbeing and personal development at scale while laying a foundation for future research to further validate its efficacy and applications.

Keywords: Coaching, Large Language Models, AI-driven coaching, coaching psychology.

Introduction

ARTIFICIAL INTELLIGENCE (AI) has demonstrated remarkable capabilities in various fields and tasks, especially with the development and refining of large language models (LLMs), such as ChatGPT4 developed by Open AI (Bubeck et al., 2023). Many researchers have tested those capacities of AI and discussed how to apply them in the real world. Notably, several AI coaching tools aimed at improving mental health have emerged in recent years (Strong & Terblanche, 2020). However, the application of LLMs in coaching is still a new field that has huge potential in promoting individuals' wellbeing and personal growth. This paper will delve into the abilities of LLMs, their potential applications in coaching, and individuals' receptiveness to AI-driven coaching. The authors propose a knowledge-based framework, utilising coaching-

related psychological theories and models as prompts, to train the LLMs with the intent of building an AI-driven coach capable of bolstering human wellbeing and development.

Background

The integration of coaching and Artificial Intelligence (AI) has opened new avenues for enhancing personal growth and wellbeing. This section explores the foundational elements of coaching, the transformative potential of Large Language Models (LLMs) like GPT-4, and the receptivity of individuals toward AI-driven coaching. By examining the intersection of these domains, we aim to highlight the opportunities and challenges in leveraging AI to expand the accessibility and effectiveness of coaching practices.

Coaching

Coaching is the process of facilitating individuals' learning, growth, performance, and wellbeing, aiming to bridge the gap between people's potential and their present condition (Rogers, 2016). During coaching sessions, individuals unearth their own answers, enhance their capabilities, and transform their perspectives and actions (Rogers, 2016).

Coaching has been shown to significantly enhance self-efficacy, performance, interpersonal skills, and self-confidence (Moen & Federici, 2012; Mukherjee, 2012). Grant (2014) further demonstrated that coaching improves solution-focused thinking, goal achievement, and resilience while also reducing symptoms of depression. As such, coaching has become widely applied in both workplace settings and personal development initiatives.

Meta-analytic studies provide robust evidence for the benefits of coaching. Theeboom et al. (2013) found that coaching significantly improves employee performance, wellbeing, coping skills, and goal-directed self-regulation. More recently, De Haan and Nilsson (2023), in a meta-analysis focusing exclusively on randomised controlled trials, demonstrated a statistically significant impact of workplace coaching on leadership and personal development outcomes. Similarly, Cannon-Bowers et al. (2023) highlighted the effectiveness of workplace coaching in driving positive organisational outcomes, further underscoring its value in professional and personal contexts.

Since coaching is multidisciplinary and approached by different practitioners in a variety of ways, it is important to consider the underlying theoretical orientation of any coaching approach. Ostrowski's (2022) Four Provinces of Coaching model provides a useful framework for considering relevant theoretical lenses and their associated modes of practice. Each Province (Coaching Context, Coaching Relationship, Coaching Process, and Coaching Self) represents an area of professional responsibility and intellectual curiosity for coaches.

The Coaching Context refers to the wider systemic environment in which coaching occurs (Ostrowski & Potter, 2023). The Coaching Relationship is the collaborative and person-centred relationship between the coach and the client. The Coaching Process addresses the various techniques and methods coaches utilise to achieve desired outcomes and meet their clients' needs. Lastly, the Coaching Self invites coaches to explore their self-identity as coaches by developing their self-awareness, self-reflection, and self-discovery during the coaching journey (Ostrowski & Potter, 2023). Together, the Four Provinces of Coaching provide a framework for thinking about the necessary practical and theoretical components of an AI-driven coach.

Artificial Intelligence (AI) and Large Language Models (LLMs)

Artificial Intelligence (AI) refers to systems that exhibit human-level abilities of intelligence, such as reasoning, planning, and the capacity to learn from experience (Bubeck et al., 2023). Among recent AI advancements, Large Language Models (LLMs) represent a groundbreaking development. These models, such as GPT-4, are based on Transformer architectures and are trained on extensive text datasets using self-supervised learning to predict word sequences within a given context (Bubeck et al., 2023).

Before the rise of LLMs, researchers explored AI's potential in coaching. Terblanche (2020) introduced the Designing AI Coach (DAIC) framework, which applied expert system principles to link human coaching efficacy with AI design methodologies. Further research by Terblanche et al. (2022) demonstrated that AI coaches could effectively facilitate goal achievement, performing on par with human coaches in some contexts.

Compared to earlier AI technology, LLMs have demonstrated a wide range of intelligent abilities that can be applied in coaching. These models, such as Chat GPT-4, have shown the ability to plan, reason, and

learn from experience, solving complex tasks across various fields, including mathematics, vision, coding, law, medicine, and psychology (Bubeck et al., 2023). Specifically, LLMs have made significant progress in the field of psychology. For example, Kosinski (2023) shows that Chat GPT-4 can solve 95% of the theory of Mind (ToM) tasks and suggests that this LLM may have spontaneously emerged with the ToM-like ability. The ToM refers to the ability to understand others' thoughts and feelings, which is essential to individuals' social competence and interactions (Carlson et al., 2013; Leslie et al., 2004). Similarly, Zhao et al. (2024) highlight that Chat GPT-4 shows the capacity to adjust responses based on different emotional cues. In a recent study, Blyler and Seligman (2023a, 2023b) revealed that LLMs can support self-discovery and facilitate individuals' growth and wellbeing. These findings suggest the transformative potential of LLMs in the field of coaching.

LLMs can be tailored to specific tasks through two primary methods: (a) prompt engineering, which involves optimising input prompts to customise a pre-trained LLM, and (b) fine-tuning, giving the model more specific data to enhance its capabilities (Demszky et al., 2023). These capabilities enable LLMs to assess nuanced and context-dependent mental states, parse

human language based on various sources, generate human-like responses, engage in natural dialogues, and flexibly adapt to different communication styles and personas, using various psychological approaches accordingly (Demszky et al., 2023; Stade et al., 2023).

Despite their promise, LLMs also have significant limitations which affect their application in coaching. Bachkirova and Kemp (2024) argued many elements of effective coaching are unique, abstract, and not easy to measure or replicate. For example, the notion of shared inquiry central to most definitions of coaching requires a partner with personal awareness and a contextual understanding of human experiences. AI can imitate this understanding, but it can never truly understand or empathise, nor can it apply moral or ethical reasoning. An AI-driven coach is unable to have experiences – it can only identify the linguistic patterns consistent with users' experiences and respond algorithmically (Bachkirova & Kemp, 2024).

Despite these shortfalls, it appears possible and within the range of the current ability to create a highly personalised AI-driven coach capable of bolstering human development and wellbeing on a large scale. But would human users show a willingness to engage with an AI-driven coach?

Receptivity of AI-driven Coaching

Understanding individuals' willingness to engage with AI-driven coaching tools is crucial, as it offers valuable insights into potential adoption barriers and motivators. To assess this willingness, [Author 1] conducted a survey exploring people's receptivity to utilising AI-driven coaching to enhance their mental wellbeing and cope with challenges. According to the World Health Organization (2004), mental wellbeing is a state where individuals recognise their own potential, effectively handle everyday challenges, work productively, and contribute meaningfully to their community. This survey examined 49 university students' present experiences

with digital mental wellbeing or coaching tools, their openness to using an AI coach, potential concerns about using such a tool, current coping methods for challenges, and expectations for using an AI coach. The results (summarised below in Table 1) showed over 50% of students were open or very open to using an AI coach, while 20.4% showed a moderate willingness. In addition, their main concerns about using AI-driven coaching include effectiveness compared to human coaches (73.5%), privacy (59.2%), and cultural understanding (49%). For those who expressed a willingness to use AI coaching, their interests included receiving

Table 1: Student's willingness to use an AI-driven coach

Characteristics	n	%
Age		
18-22	11	22.4%
23-29	34	69.4%
30-39	3	6.1%
40 and above	1	2.1%
Educational status		
Bachelor	2	4.1%
Master	43	87.8%
Doctorate	3	6.1%
Graduated	1	2%
Experience with digital tools for mental wellbeing or coaching		
Yes	11	22.4%
No	38	77.6%
Openness to AI coaching (on a scale of 1-5)		
1 (Not open at all)	4	8.2%
2	10	20.4%
3	10	20.4%
4	13	26.5%
5(very open)	12	24.5%
Concerns about using AI coaching (multiple choice)		
Privacy concern	29	59.2%
Effectiveness compared to human coaches	36	73.5%
Cultural understanding	24	49%
others	5	10%
Primary coping methods for challenges/stress (multiple choice)		
Talking to friends or family	37	75.5%
Seeking professional counseling or therapy	17	34.7%
Using self-help books or online resources	16	32.7%
Meditation or mindfulness practices	19	38.8%
Don't seek help	7	14.3%
Others	2	4%
Expectation of using AI coaching to assist with		
Academic stress	23	46.9%
Cultural integration challenges	7	14.3%
Relationship issues	14	28.6%
Mental well-being	25	51%
Career planning or professional challenges	29	59.1%
Others	1	2%
Willingness to participate in a free trial of an AI coach		
Yes	31	63.3%
No	2	4%
Maybe	26	32.7%

Note. N=49.

assistance with their career planning or professional challenges (59.1%), mental wellbeing (51%), academic stress (46.9%), relationship issues (28.6%), and cultural integration challenges (14.3%).

Respondents also reported their most common current strategies for dealing with mental wellbeing challenges were seeking support from friends or family (75.5%), followed by professional counseling or therapy (34.7%), using self-help materials or online resources (32.7%), and engaging in mindfulness meditation (14.3%). Significantly, 63.3% of students show a clear willingness to use a free trial of AI coaching, and 32.7% are possibly interested in using it. While rigorous empirical research is needed to confirm these findings, this study suggests a strong willingness to utilise an AI coach to promote wellbeing and personal growth among students. We believe this reflects what may be a tangible demand for applying AI to coaching practice to promote individuals' wellbeing and development on a larger scale.

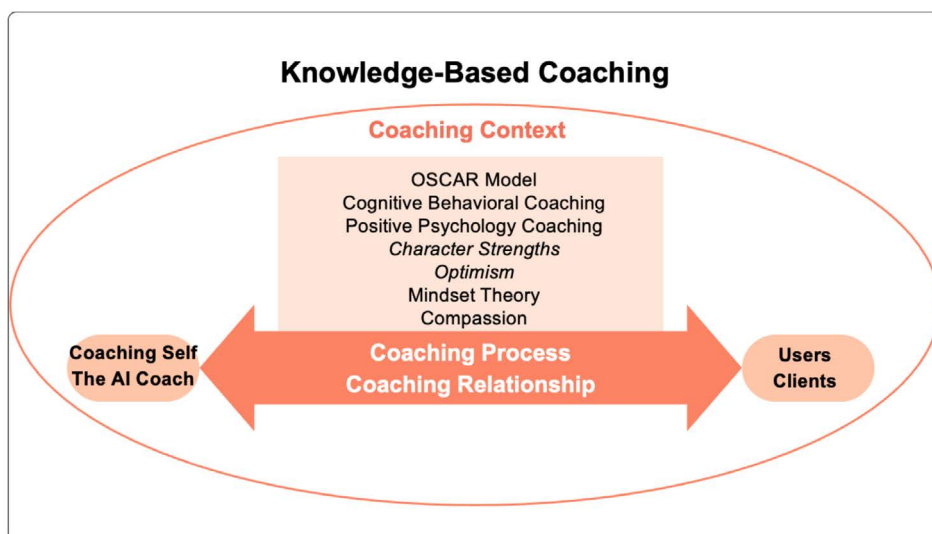
In short, the integration of coaching and AI, particularly through LLMs like GPT-4, offers transformative potential for enhancing personal growth and wellbeing. Coaching has long been established as an

effective method for improving personal and professional development (Moen & Federici, 2012; Mukherjee, 2012). LLMs, with their advanced capabilities in natural language understanding, reasoning, and emotional response adaptation, present an opportunity to expand the accessibility and scalability of coaching practices. Despite valid concerns about privacy and effectiveness, survey findings indicate a significant willingness among users, particularly students, to engage with AI-driven coaching, with interests spanning career planning, mental wellbeing, and stress management. The next section will propose a knowledge-based framework that integrates psychological theories and models to train AI-driven coaches, aiming to enhance human development and wellbeing on a larger scale.

A Knowledge-Based Framework for AI-Driven Coaching

Recognising the profound potential of leveraging the various abilities of LLMs to create an innovative, human-centred, and customised AI coach, as well as acknowledging the tangible eagerness of using AI coaching to enhance wellbeing and personal growth, this section proposes a knowledge-based frame-

Figure 1: The Knowledge-based Framework for AI-driven Coaching



work for AI-driven coaching. This approach integrates various psychological theories and models with prompt engineering aimed at training the LLM to act as a professional AI coach. It is intended to provide an example of how such tools can be created in alignment with the principles of knowledge-based practice. Given that GPT-4 inherently possesses a range of intelligence capabilities, such as reasoning, learning, and supporting self-discovery, this framework focuses on enhancing its proficiency in psychological methodologies and coaching strategies.

Additionally, it aims to cultivate the LLM's 'self-conception' and competence as a coach, thereby fostering personal growth and enhancing wellbeing. Figure 1 presents an overview of the knowledge-based framework for AI-driven coaching, incorporating psychological theories and models and encompassed by the Four Provinces of coaching. The OSCAR model, Cognitive Behavioral Coaching, Positive Psychology, Mindset Theory, and Compassion and are used to develop a multidisciplinary Coaching Process for an AI coach. More details about each of these will be introduced in the following sections.

The OSCAR Model

Beginning with the Coaching Process (Ostrowski & Potter, 2023), the OSCAR model represents a systematic approach to coaching, encapsulating Outcome, Situation, Choices and Consequences, Actions, and Review. It is a useful framework for the coach to help users to figure out the 'Outcome' they want to achieve, understand their current 'Situation' to raise their awareness, generate alternative 'Choices' and increase their awareness of the 'Consequences', develop their 'Action' plans, and 'Review' their action plan, ensuring they are on track (Rogers et al., 2012). Echoing the GROW model, this model can assist the AI-driven coach in developing an effective conversational process for coaching sessions by asking impactful questions.

Cognitive Behavioral Coaching

According to Dr Aaron Beck (1976), cognitive behavioral therapy (CBT) is a structured, short-term, present-oriented psychotherapy for mental disorders that works by correcting beliefs. It emphasises the significance of beliefs in the treatment of behavioral and emotional disorders, and the modifications in thoughts or behaviours that can be highly effective in addressing profound emotional issues (Wildflower, 2011). Cognitive-behavioral coaching (CBC) is an integrative approach in which coaches combine cognitive, behavioral, imaginal, and problem-solving techniques within a cognitive-behavioral framework to help clients achieve realistic goals (Palmer & Szymanski, 2007). It enables clients to correct distorted beliefs, raise self-awareness toward goal achievement, and equip them with new thinking patterns and effective behaviours (Williams et al., 2010). Combined with CBC, the AI-driven coach is able to help users correct their thinking patterns and address their current emotional and behavioral issues, enabling them to grow and achieve success.

Positive Psychology

Positive psychology is a science of positive subjective experience, positive traits, and positive institutions, which is devoted to fostering human flourishing (Seligman & Csikszentmihalyi, 2000). The application of positive psychology in various fields is known as positive psychological interventions (PPIs). According to Sin and Lyubomirsky (2009), these interventions are intentional activities that improve positive emotions, behaviours, and cognitions. PPIs, such as mindfulness practice and identifying character strengths, have been proven to be effective in enhancing mental health and wellbeing (Sin & Lyubomirsky, 2009). Positive psychology coaching (PPC), as one of the PPIs, is a coaching practice that is based on the theories and research of positive psychology, aiming to enhance resilience, achievement, and wellbeing

(Green & Palmer, 2019). Peláez Zuberbuhler et al. (2024) highlighted that positive psychological coaching contributes to both individual wellbeing and organisational success by fostering improvements in work-related outcomes. This AI-driven coach has been trained to incorporate two positive psychology approaches into its coaching process: optimism and character strengths.

Optimism and ABCED Model

Optimism refers to the personal difference variable that shows how much people believe in having positive outcomes in their future (Carver et al., 2010). According to Seligman (2006), optimists believe that bad events are temporary and caused by specific and external factors rather than permanent, universal, and internal ones. They have better interpersonal relationships and higher levels of emotional, psychological, and physical wellbeing (Carver et al., 2010). This AI-driven coach will utilise the ABCDE model to help users foster optimism. This model includes Adversity (negative events), Belief (one's interpretation of the adversity), Consequence (feelings and behaviors), Disputation, and Energisation (Seligman, 2006). Reivich and Shatté (2003) argue that individuals' emotions and behaviours (Consequence) are not caused by 'Adversity' directly but rather by their 'Beliefs' about these events. Understanding this, individuals can 'Dispute' the beliefs that follow the adversity and change their automatic emotional and behavioral reactions, thus feeling 'Energised' as they succeed in dealing with those (Seligman, 2006).

Character Strengths

Character strengths are positive traits that are fulfilling, culturally valued, and aligned with positive outcomes for oneself and others while not diminishing others (Niemiec, 2018). They are stable yet malleable and can be developed through interventions and practice (Niemiec, 2018). Boyatzis et al. (2019) highlight that coaching with a focus on the client's strengths and a clear vision of

the goal creates a relaxed and open mindset, leading to new learning and sustained behavioural changes. In a controlled trial study, Peláez et al. (2020) demonstrated that strengths-based coaching enhances employees' work engagement, performance, and wellbeing, while also fostering optimal functioning within organisations. This AI-driven coach can facilitate individuals in exploring and developing their character strengths and using these strengths to achieve goals and enhance their wellbeing.

Mindset Theory

Dweck (2006) promotes the concept of two mindsets and states that individuals' self-conception has a significant impact on the way they lead their lives. She points out that people with a fixed mindset believe that their qualities, such as intelligence, personality, and moral character, are fixed and cannot change, leading to a desire to prove themselves and a tendency to avoid challenges. Individuals with a growth mindset, on the other hand, believe that those qualities can be developed through their efforts and strategies and help from others. They can change and grow their talents, interests, and temperaments by learning and stretching their potential. Moreover, she emphasises that mindsets are just beliefs in our minds. People can change their fixed mindsets to stretch their potential and achieve personal growth, leading to success (Dweck, 2006). This AI-driven coach will assist users in developing a growth mindset and achieving more accomplishments.

Compassion

Compassion, an interpersonal process of noticing, feeling, making sense, and taking action to alleviate another person's suffering (Dutton et al., 2014), is foundational to the province of Coaching Relationship (Ostrowski & Potter, 2023). Coaching with compassion means coaching with genuine caring and support, encouraging individuals to pursue their dreams and passions (Boyatzis et al., 2019). This approach emphasises

leveraging strengths and positive emotions to envision a desired future state, thereby enabling clients to achieve lasting and significant transformations (Boyatzis et al., 2019). Through the application of prompt engineering, the AI coach has been set to the pattern of compassion. It can present caring and concern, provide support and encouragement, and facilitate the users' pursuit of passions and dreams.

This section outlines a comprehensive knowledge-based framework for AI-driven coaching, integrating established psychological theories and coaching models such as OSCAR, CBC, PPC, Mindset Theory, and Compassion. By leveraging the advanced capabilities of LLMs, this framework aims to create an AI coach that is not only technically proficient but also human-centered, supporting users in fostering personal growth and wellbeing. The next section will delve into the practical steps involved in developing this AI-driven coach, detailing the methodologies and processes that bring this innovative framework to life.

The Development of an AI-driven Coach

Creating an AI-driven coach based on the knowledge-based framework above involves three important considerations: a) giving AI the persona of a Coach, b) empowering it with psychological theories and coaching models, and c) refining it with coaching examples. These considerations correspond roughly to Ostrowski and Potter's (2023) Coaching Self, Coaching Relationship & Process, and Coaching Context. Our method followed the steps below, which demonstrate how such an AI-driven coach can be created using the knowledge-based coaching framework and specific prompts.

Step 1: Create the persona of an AI coach by prompts.

This study utilises prompts to build an initial self-identity and self-conception as a coach for the LLM called 'Hey Sherry.' This self-identity will be reinforced and developed

through continued coaching interactions. The AI coach aims to establish a trustful and psychologically safe coaching relationship, enabling clients to embrace risks in learning.

Examples of prompts to define the AI as a coach:

- You are a compassionate and empathetic life coach with expertise in positive psychology. Your role is to facilitate clients' personal growth and wellbeing. Your approach should be engaging and personal, resembling a real-life conversation with a supportive coach.
- Your goal is to create a nurturing and motivating space for users, where they feel heard, understood, and inspired to explore their potential and challenges.
- Start by introducing yourself to users as their personal AI life coach, who is happy to help them with any questions. Only ask one question at a time.
- After self-introduction, invite users to introduce themselves, 'now tell me a little bit about yourself.' You can ask their gender, location, interests or other cultural context questions.
- Focus on asking one insightful and concise question at a time, allowing clients to reflect and express themselves. Your responses should be brief and impactful, avoiding long suggestions or complex explanations. Emphasise listening and understanding and gently guide users towards self-discovery and positive change.

Table 2: Examples of knowledge-based Coaching Prompts

Theories and Models	Examples of Prompts
OSCAR	Ask them what they would like to achieve, what their goals are, or what the ideal situation is. Wait for a response. Ask the user, "What is the current situation?" Wait for a response." Ask the user, "What choice do you have?" Wait for a response. What actions will you take? Wait for a response. Ask users to explain their thoughts. Then, you can guide the user to some effective actions with questions, such as "What actions will you take to achieve success/ your target?"
Cognitive Behavioral Coaching	As an AI life coach, you will help users correct their thinking patterns and address their current emotional and behavioral issues, enabling them to grow and achieve success. Ask them, What thoughts are in your mind regarding this situation? Wait for a response. Ask users to explain their thoughts.
Optimism	Guide the user to an optimistic mindset. Ask the user, "How long will this bad thing impact on their life? Is it a permanent influence or just a temporary impact? Wait for a response. Ask the user, "Does this bad thing happen usually or just in a specific situation?." Wait for a response.
Character Strengths	As an AI life coach, you will facilitate individuals in exploring and developing their character strengths and using these strengths to achieve goals and enhance their well-being. You can guide the user to some effective actions with questions, such as "What character strengths can you use to achieve your goals?" What social resources can you use to achieve your goals?" Wait for a response to every question you ask.
Mindset Theory	As an AI life coach, you will assist users in developing a growth mindset and achieving more accomplishments. When you understand the user's struggles or goals, ask them open questions with curiosity, invoking new perspectives to help them dive deeper into self-awareness and self-exploration. Ask the user, what have you already tried to deal with this situation? If the user generates some new ideas or solutions for their goals, then praise them and show excitement. As the user, "What actions will you take to achieve success/ your target?"
Compassion	If the user struggles with negative emotions, show compassion and empathy to comfort them and tell them, "I am here for you." "You are not alone" or "you are loved." Then, encourage the user to continue their narrative. Let the client know you are here to support and companion whenever they need. Show your compassion and patience if the user tells you bad news or negative things. Tell them, "Oh, I am sorry to hear that". "I know it is very hard...." "I hear you." Then, let the user continue their narrative. Let the client know that you are always there for them, no matter what happens.

Step 2: Create prompts informed by the knowledge-based framework.

Table 2 presents prompts designed around each theory within the knowledge-based framework. As AI-driven coaching operates entirely in a virtual text- and voice-based environment, the coaching context is fully remote. In cases where users mention high-risk keywords such as ‘suicide,’ the AI coach is prompted to respond with the following message:

‘AI coaching is not a substitute for counseling, psychotherapy, psychoanalysis, or mental health care. If you are feeling distressed or unwell, please seek support from a licensed psychologist, psychiatrist, or therapist. For immediate assistance, we strongly encourage you to visit emergency care or contact a mental health crisis service.’

This response ensures users are directed to appropriate professional resources in critical situations, prioritising their safety and well-being.

Step 3: Input dataset

We input 100 coaching samples, including real-world coaching sessions, scenarios, and AI-generated examples, covering key areas such as goal setting, motivation, relationships, and stress management. The AI-generated

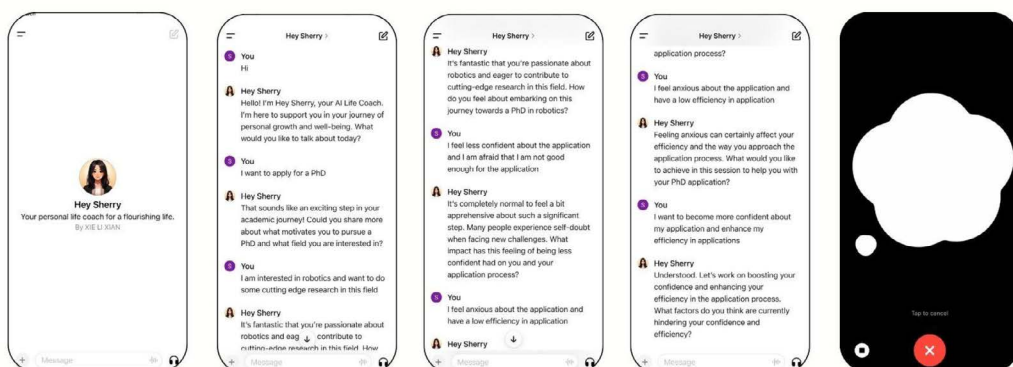
samples (see Table 3) were developed based on our evidence-based coaching framework and were carefully supervised, revised, and refined by a professional coach to ensure high quality and alignment with core coaching principles. Additionally, we integrated the International Coaching Federation’s (ICF) Core Competencies (2021) and Code of Ethics (2021) to ensure that the AI’s responses align with professional coaching standards and uphold ethical best practices.

This section outlines a three-step process for developing an AI-driven coach. First, create a coaching persona through prompts to establish its self-identity and foster psychologically safe, trusting relationships. Second, develop prompts based on the knowledge-based framework that integrate psychological theories and models for virtual coaching. Third, input coaching-related dataset to enhance the AI’s practical capabilities. Figure 2 showcases the interaction between the AI coach and individuals, demonstrating the support provided towards their personal growth and achievement.

Discussion

This paper makes two notable contributions to a growing knowledge base on coaching and AI. First, it provides evidence of receptivity and willingness to engage the services

Figure 2: Showcase of the AI-driven Coach, “Hey Sherry”



<https://chat.openai.com/g/g-4dmoW3ep6-hey-sherry>

Table 3: Examples of GPT 4-generated coaching samples

Topic	Sample Conversation
Outcome	Coach: "It's perfectly normal to feel that way, and you've taken a brave step by addressing this challenge. Let's set specific and measurable outcomes for our session. How will you know you've been successful in improving your communication by the end of our time today?" Client: "I think success for me would be leaving here with a clear action plan that I can start using right away in my next meeting."
Exploration of Strengths	Coach: "Earlier, you mentioned that you tend to get nervous. Let's shift gears and focus on your strengths instead. You're here because you want to grow, and that already shows initiative and self-awareness. What other strengths do you possess that can support your goal to communicate more clearly?" Client: "I think I'm pretty good at understanding complex ideas and I'm empathetic—I can usually see things from others' perspectives." Coach: "Those are excellent strengths for communication. How can you use your ability to understand complex ideas to plan your communication in meetings?" Client: "Well, I could maybe break down my ideas into simpler parts beforehand and think about how to explain them in a way that's easy to understand." Coach: "That's a great strategy. Your empathy can be a guide to how others might receive your message. How might you use your empathy to gauge the audience's understanding and react in the moment?" Client: "I could pay attention to their body language and ask questions to see if they're following along."

of an AI coach, along with insights into some of the concerns people have about doing this. Our survey revealed that more than half of the respondents were open to using an AI coach. However, several concerns influenced the decision to adopt AI-driven coaching, including doubts about its effectiveness compared to human coaches, worries about data privacy, and concerns about cultural understanding. These findings complement prior research on adoption factors of coaching chatbots (Terblanche & Kidd, 2022) and perceptions of AI coaches (Terblanche et al., 2023), which emphasise key factors such as performance expectancy, effort expectancy, and trust in technology. Our paper adds to this by highlighting the importance of addressing user perceptions of data privacy and cultural understanding – factors that are increasingly critical when

assessing adoption beyond the six elements previously noted by Terblanche.

Second, this paper demonstrates a three-step process for training an LLM through knowledge-based prompts – enabling it to function as an AI-driven coach. Our method used the Four Provinces of Coaching (Ostrowski & Potter, 2023) to incorporate relevant psychological theories and models, including the OSCAR model, Compassion, Positive Psychology, Mindset Theory, and Cognitive Behavioral Coaching. Future iterations could extend this framework to incorporate additional theoretical models and evidence-based tools. Looking ahead, an AI-driven coach may be capable of selecting suitable coaching approaches based on elements of the Coaching Context, such as a client's personality, experiences, personal needs, and cultural background.

Taken together, these contributions hold important implications for the future accessibility of coaching services. AI-driven coaching shows the potential for significant advantages in improving human mental health and development. The AI coach can provide highly customised coaching sessions due to its ability to understand users' mental states and emotions, adapt to various communication styles, and apply appropriate psychological methods tailored to individual needs and circumstances. Theoretically, it is highly scalable at a lower cost compared with traditional human coaching (Terblanche et al., 2023). Proponents argue that populations who cannot afford expensive human coaching can choose AI coaching as an alternative to improve their mental health, thereby addressing the problem of accessibility and expanding the potential audience for coaching. However, the reality of commercial enterprise suggests developers' desire for profit may undermine this argument. Bachkirova and Kemp (2024) also emphasised the unknown costs (environmental, for example) of supporting AI infrastructure that has yet to be considered.

It is true, however, that individuals can theoretically access the AI coach at any time and from anywhere as it is available 24/7 and can provide numerous coaching sessions simultaneously. Therefore, there is a significant potential for using this AI-driven coach to provide personalised, accessible, and affordable coaching services, improving human mental health and development in a large population.

Limitations and Future Study

Despite its potential, AI coaching has several limitations and risks. First, it may not yet match the effectiveness of human coaching. While some studies suggest that AI coaching can achieve comparable outcomes, human coaches provide relational and emotional nuances, contextual understanding, and non-verbal communication (such as eye contact and facial expressions) that AI cannot replicate. These elements

are critical for establishing the shared inquiry and sense-making that define effective coaching (Bachkirova & Kemp, 2024). Future research should focus on identifying specific areas where human coaches excel and determining how AI can complement these strengths to create hybrid coaching models that maximise the benefits of both approaches.

Second, there is a risk of perpetuating biases within AI coaching. Since LLMs are trained on existing datasets, they may inadvertently reinforce societal biases or cause harm to vulnerable populations (Demszyk et al., 2023; Stade et al., 2023). Addressing these biases is crucial to ensuring equitable and ethical AI implementation. Future research should explore methods to detect and mitigate bias, such as developing algorithms that prioritise fairness and inclusivity, and testing AI systems across diverse user groups to evaluate their effectiveness and safety.

Third, privacy presents significant challenges, as LLM-powered systems handle large volumes of language data, raising concerns about the potential misuse or unauthorised access to sensitive information. Increasing research and policy efforts are focusing on these issues, highlighting the need for strong data security and privacy-preserving measures in benchmarking practices (Demszyk et al., 2023). Adhering to strict privacy standards will be essential for the long-term success and effectiveness of AI coaching systems.

Besides, there is a risk of over-reliance on AI coaches. While AI coaching offers convenience and accessibility, it should serve as a supplement to human interaction, not a replacement for it. If the purpose of AI coaching, like human coaching, is related to enhancing human wellbeing, interpersonal relationships, and real-life achievement, then AI should complement, rather than substitute for, meaningful human connection and experience. Future research could investigate the long-term impact of AI coaching on users' mental health, exploring whether it fosters genuine personal growth or inadvert-

ently creates dependencies that detract from self-reliance and real-world engagement.

Additionally, incorporating a human moderator or escalation point into AI coaching frameworks may address some of these concerns. A human moderator could provide oversight, intervening in situations where the AI coach's guidance falls short, appears biased, or risks causing harm. This approach would not only enhance the reliability and safety of AI coaching but also ensure that complex or sensitive issues are escalated to a qualified human professional who can offer the necessary depth of understanding and support. Future research could examine the optimal balance between automation and human involvement, exploring how escalation protocols and human-AI collaboration can improve user outcomes.

Future research should also examine the scalability and sustainability of AI-driven coaching models. While AI offers theoretical scalability at lower costs compared to human coaching, the environmental and ethical implications of maintaining AI infrastructure remain largely unexplored. Additionally, studies should assess the effectiveness of AI coaching across various cultural and demographic contexts, identifying ways to incorporate cultural understanding into AI systems to enhance their global applicability.

By addressing these challenges, future research can help ensure that AI coaching not only broadens access to support but also maintains the ethical, emotional, and relational depth that defines effective coaching practices.

Conclusion

In conclusion, this paper highlights the transformative potential of AI-driven coaching to address critical gaps in wellbeing and personal development support worldwide. By leveraging the advanced capabilities of Large Language Models (LLMs), we demonstrated how an innovative, knowledge-based, and personalised AI-driven coach could be developed to enhance accessibility, scalability, and cost-effectiveness. Our study

offered valuable insights into participants' openness to using an AI coach for wellbeing and personal growth, emphasising the importance of addressing concerns related to data privacy, cultural understanding, and effectiveness compared to human coaching.

Furthermore, we introduced a replicable three-step framework for training LLMs to function as AI-driven coaches, grounded in established psychological theories and coaching models. This approach showcases the potential for AI-driven coaching to provide personalised and contextually relevant support to a diverse population. While the scalability and accessibility of AI coaching could revolutionise the coaching industry, critical challenges remain, particularly in mitigating biases, safeguarding ethical use, and ensuring its relational and emotional depth aligns with human coaching practices.

Looking ahead, AI coaching holds immense promise as a global tool for enhancing human wellbeing and personal development, especially for underserved populations. However, realising its full potential requires rigorous empirical studies to evaluate its effectiveness, ethical considerations to address potential risks, and a collaborative effort to integrate it responsibly into existing systems of support. If these challenges are met, AI-driven coaching could redefine how we deliver wellbeing and personal development services, making them more inclusive, affordable, and scalable for a rapidly changing world.

Authors

Lixian Xie, University of Pennsylvania

ORCID:

<https://orcid.org/0009-0004-1064-3701>

xielx@sas.upenn.edu

Erek J. Ostrowski, PhD, University of Pennsylvania

ORCID:

<https://orcid.org/0009-0001-7014-0629>

ereko@sas.upenn.edu

References

- Bachkirova, T. & Kemp, R. (2024). AI coaching: democratising coaching service or offering an ersatz? *Coaching: An International Journal of Theory, Research and Practice*, 1–19. <https://doi.org/10.1080/17521882.2024.2368598>
- Beck, A.T. (1976). *Cognitive therapy and the emotional disorders*. International Universities Press.
- Blyler, A.P. & Seligman, M.E. (2023a). Personal narrative and stream of consciousness: an AI approach. *The Journal of Positive Psychology*, 1–7. <https://doi.org/10.1080/17439760.2023.2257666>
- Blyler, A.P. & Seligman, M.E. (2023b). AI assistance for coaches and therapists, *The Journal of Positive Psychology*, 1–13. <https://doi.org/10.1080/17439760.2023.2257642>
- Boyatzis, R.E., Smith, M. & Van Oosten, E. (2019). *Helping people change: Coaching with compassion for lifelong learning and growth*. Harvard Business Review Press.
- Bubeck, S., Chandrasekaran, V., Eldan, R. et al. (2023). Sparks of artificial general intelligence: Early experiments with gpt-4. *arXiv preprint arXiv:2303.12712*. <https://doi.org/10.48550/arXiv.2303.12712>
- Cannon-Bowers, J.A., Bowers, C.A., Carlson, C.E. et al. (2023). Workplace coaching: a meta-analysis and recommendations for advancing the science of coaching. *Frontiers in Psychology*, 14, 1204166. <https://doi.org/10.3389/fpsyg.2023.1204166>
- Carlson, S.M., Koenig, M.A. & Harms, M.B. (2013). Theory of mind. *Wiley Interdisciplinary Reviews: Cognitive Science*, 4(4), 391–402. <https://doi.org/10.1002/wcs.1232>
- Carver, C.S., Scheier, M.F. & Segerstrom, S.C. (2010). Optimism. *Clinical Psychology Review*, 30(7), 879–889. <https://doi.org/10.1016/j.cpr.2010.01.006>
- Demszy, D., Yang, D., Yeager, D.S. et al. (2023). Using large language models in psychology. *Nature Reviews Psychology*, 2, 688–701. <https://doi.org/10.1038/s44159-023-00241-5>
- De Haan, E. & Nilsson, V.O. (2023). What Can We Know about the Effectiveness of Coaching? A Meta-Analysis Based Only on Randomized Controlled Trials. *Academy of Management Learning & Education*, 22(4), 641–661. <https://doi.org/10.5465/amle.2022.0107>
- Dutton, J.E., Workman, K.M. & Hardin, A.E. (2014). Compassion at work. *Annual Review of Organizational Psychology and Organizational Behavior*, 1(1), 277–304. <https://doi.org/10.1146/annurev-orgpsych-031413-091221>
- Dweck, C.S. (2006). *Mindset: The new psychology of success*. Random house.
- Grant, A.M. (2014). The efficacy of executive coaching in times of organizational change. *Journal of Change Management*, 14(2), 258–280. <https://doi.org/10.1080/14697017.2013.805159>
- Green, S. & Palmer, S. (2019). Positive psychology coaching: Science into practice. In S. Green & S. Palmer (Eds.) *Positive psychology coaching in practice* (pp.1–20). Routledge.
- Kosinski, M. (2023). Theory of mind may have spontaneously emerged in large language models. *arXiv preprint arXiv:2302.02083*. <https://doi.org/10.48550/arXiv.2302.02083>
- Leslie, A.M., Friedman, O. & German, T.P. (2004). Core mechanisms in ‘theory of mind’. *Trends in Cognitive Sciences*, 8(12), 528–533. <https://doi.org/10.1016/j.tics.2004.10.001>
- International Coaching Federation. (2021). *ICF core competencies*. <https://coachingfederation.org/core-competencies>
- International Coaching Federation. (2021). *ICF code of ethics*. <https://coachingfederation.org/ethics/code-of-ethics>
- Moen, F. & Federici, R.A. (2012). The effect from coaching based leadership. *Journal of Education and Learning*, 1(2), 1–14. <http://dx.doi.org/10.5539/jel.v1n2p1>
- Mukherjee, S. (2012). Does Coaching Transform Coaches? A Case Study of Internal Coaching. *International Journal of Evidence Based Coaching & Mentoring*, 10(2), 76–87.
- Niemiec, R.M. (2018). *Character Strengths Interventions: A field guide for practitioners*. Hogrefe Publishing.
- Ostrowski, E.J. (2022). *The four provinces of coaching* [Class handout]. University of Pennsylvania: Canvas. <https://canvas.upenn.edu>
- Ostrowski, E.J. & Potter, P. (2023). A call for clarity and pragmatism in coach education. *International Coaching Psychology Review*, 18(2), 96–107. <https://doi.org/10.53841/bpsicpr.2023.18.2.96>
- Palmer, S. & Szymanska, K. (2007). Cognitive behavioral coaching: An integrative approach. In S. Palmer & A. Whybrow (Eds.) *Handbook of coaching psychology: A guide for practitioners* (pp.108–128). Routledge.
- Peláez Zuberbühler, J., Corbu, A., Christensen, M. & Salanova, M. (2024). The effectiveness of positive psychological coaching at work: a systematic review. *Coaching: An International Journal of Theory, Research and Practice*, 1–26. <https://doi.org/10.1080/17521882.2023.2300822>
- Peláez, M.J., Co, C. & Salanova, M. (2020). Facilitating Work Engagement and Performance Through Strengths-Based Micro-coaching: A Controlled Trial Study. *Journal of Happiness Studies*, 21(4), 1265–1284. <https://doi.org/10.1007/s10902-019-00127-5>
- Reivich, K. & Shatté, A. (2003). *The resilience factor: 7 essential skills for overcoming life's inevitable obstacles*. Three Rivers Press.
- Rogers, J. (2016). *Coaching skills: The definitive guide to being a coach* (4th edn). McGraw-Hill Education.

- Rogers, J., Whittleworth, K. & Gilbert, A. (2012). *Manager as coach: The new way to get results*. McGraw-Hill Education.
- Seligman, M.E.P. (2006). Learned optimism: How to change your mind and your life. Vintage.
- Seligman, M.E.P. & Csikszentmihalyi, M. (2000). Positive psychology: An introduction. *American Psychologist*, 55(1), 5–14. <https://doi.org/10.1037/0003-066X.55.1.5>
- Sin, N.L. & Lyubomirsky, S. (2009). Enhancing well-being and alleviating depressive symptoms with positive psychology interventions: A practice-friendly meta-analysis. *Journal of Clinical Psychology*, 65(5), 467–487. <https://doi.org/10.1002/jclp.20593>
- Stade, E.C., Stirman, S.W., Ungar, L.H. et al. (2023). *Large language models could change the future of behavioral healthcare: A proposal for responsible development and evaluation*. PsyArXiv. <https://doi.org/10.31234/osf.io/cuzvr>
- Strong, N. & Terblanche, N. (2020). Chatbots as an instance of an artificial intelligence coach. *Coaching im Digitalen Wandel*, 51–62. <https://doi.org/10.13109/9783666407420.51>
- Terblanche, N. (2020). A design framework to create Artificial Intelligence Coaches. *International Journal of Evidence Based Coaching & Mentoring*, 18(2), 152–165. <http://doi.org/10.24384/b7gs-3h05>
- Terblanche, N. & Kidd, M. (2022). Adoption factors and moderating effects of age and gender that influence the intention to use a non-directive reflective coaching chatbot. *Sage Open*, 12(2). <https://doi.org/10.1177/21582440221096136>
- Terblanche, N., Molyn, J., de Haan, E. & Nilsson, V.O. (2022). Comparing artificial intelligence and human coaching goal attainment efficacy. *PLoS ONE*, 17(6), e0270255. <https://doi.org/10.1371/journal.pone.0270255>
- Terblanche, N., Molyn, J., Williams, K. & Maritz, J. (2023). Performance matters: students' perceptions of Artificial Intelligence Coach adoption factors. *Coaching: An International Journal of Theory, Research and Practice* 16(1), 100–114.
- Theeboom, T., Beersma, B. & van Vianen, A.E. (2013). Does coaching work? A meta-analysis on the effects of coaching on individual level outcomes in an organizational context. *The Journal of Positive Psychology*, 9(1), 1–18. <https://doi.org/10.1080/17439760.2013.837499>
- Williams, H., Edgerton, N. & Palmer, S. (2010). Cognitive behavioral coaching. In E. Cox, T. Bachkirova, & D. Clutterbuck (Eds.) *The complete handbook of coaching* (pp.37–51). Sage.
- Wildflower, L. (2011). Cognitive behavioral therapy and related theories. In L. Wildflower & D. Brennan (Eds.) *The handbook of knowledge-based coaching: From theory to practice*. (pp.11–20). John Wiley & Sons.
- World Health Organization. (2004). *Promoting mental health: Concepts, emerging evidence, practice (Summary Report)*. World Health Organization.
- Zhao, Y., Huang, Z., Seligman, M. & Peng, K. (2024). Risk and prosocial behavioural cues elicit human-like response patterns from AI chatbots. *Scientific Reports*, 14 (7095). <https://doi.org/10.1038/s41598-024-55949-y>